

27. Do Cropping, Copying and pasting image inside another image using OpenCV.

AIM:

Cropping, Copying and pasting image inside another image using OpenCV.

PROGRAM:

1. For Cropping

```
import cv2

img = cv2.imread(r"C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png")

if img is None:
    print("Error: Image not found!")
else:
    # Crop the image from (10,10) to (300,300)
    crop_img = img[10:300, 10:300]

    # Save the cropped image
    cv2.imwrite(r"C:\Users\Susmitha\Downloads\ITA0510 Lab\cropped_image.jpg",
    crop_img)

    cv2.imshow("Cropped Image", crop_img)
    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

2. For Copy & Paste import

```
cv2

# Background image img1 =
cv2.imread(r"C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png")

# Object image (Shinchan) img2 =
cv2.imread(r"C:\Users\Susmitha\Downloads\ITA0510 Lab\exp2.png")

if img1 is None or img2 is None:
    print("Error: One or both images not found!") else:
    # Resize object if it is too large
    rows, cols = img2.shape[:2]

    if rows > img1.shape[0] or cols > img1.shape[1]:
        scale = min(img1.shape[0] / rows, img1.shape[1] / cols) * 0.4
        img2 = cv2.resize(img2, (int(cols * scale), int(rows * scale)))
    rows, cols = img2.shape[:2]
```

```

# Region of Interest    roi =
img1[50:50+rows, 50:50+cols]

# Convert object image to grayscale    img2gray =
cv2.cvtColor(img2, cv2.COLOR_BGR2GRAY)

# Create mask
_, mask = cv2.threshold(img2gray, 240, 255, cv2.THRESH_BINARY_INV)
mask_inv = cv2.bitwise_not(mask)

# Black-out the area in ROI    img1_bg =
cv2.bitwise_and(roi, roi, mask=mask_inv)

# Extract only the object    img2_fg =
cv2.bitwise_and(img2, img2, mask=mask)

# Combine    dst =
cv2.add(img1_bg, img2_fg)

# Place the result back
img1[50:50+rows, 50:50+cols] = dst

# Save output    cv2.imwrite(r"C:\Users\Susmitha\Downloads\ITA0510
Lab\copy_paste_result.jpg", img1)

# Display
cv2.imshow("Result", img1)
cv2.waitKey(0)
cv2.destroyAllWindows()

```

INPUT:

Image 1:



Image 2:



OUTPUT:



RESULT:

Thus the program is executed successfully.